Department of Engineering Science
School of Science & Technology



CES 440 Data Communications
Ch.5 The Network Layer: Part II

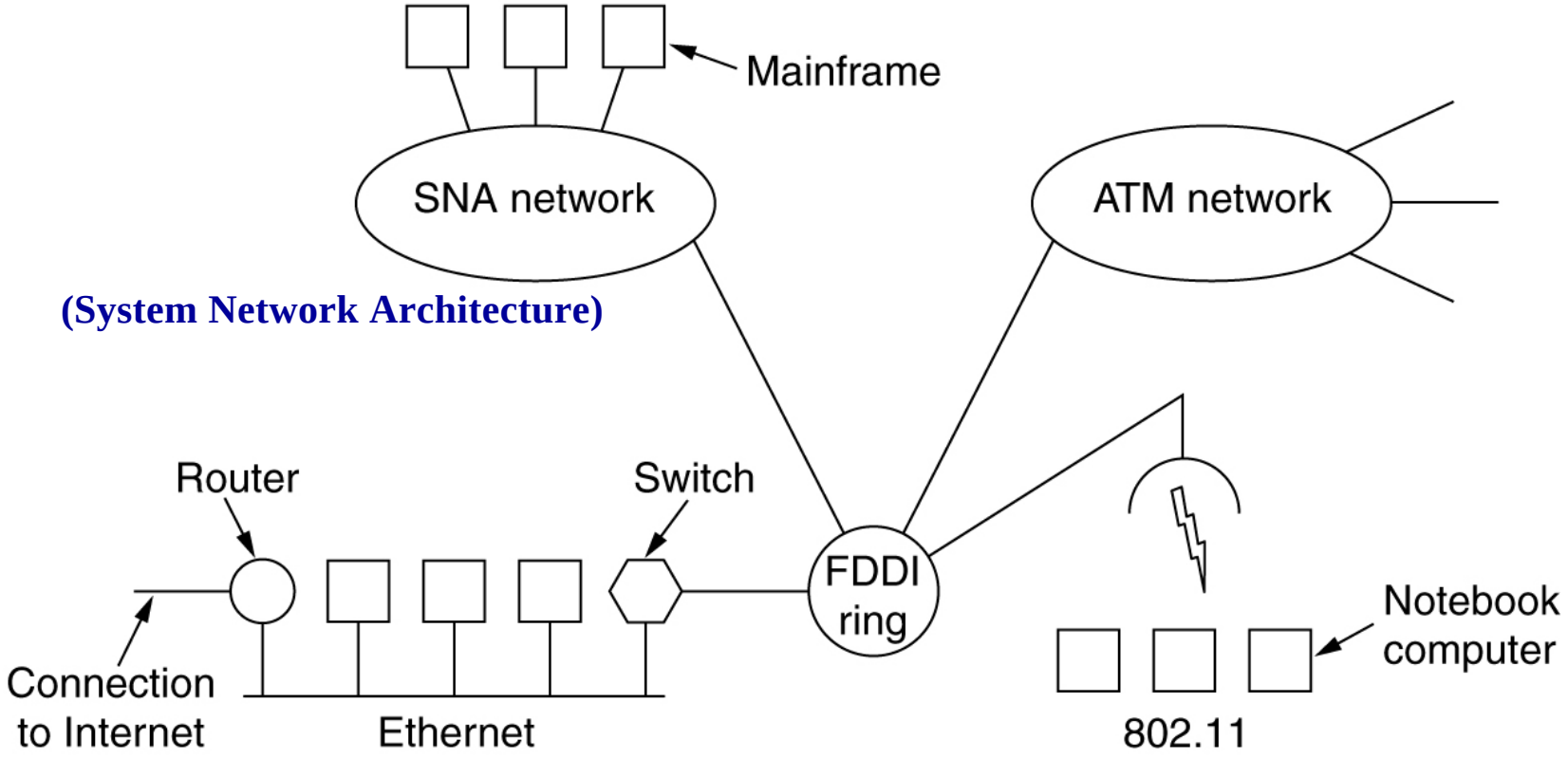
Dr. Jingxian Wu
(707) 664-3491
Jingxian.wu@sonoma.edu

OUTLINE

- Design Issues
- Routing Algorithms
- **Internetworking**
- Quality of Service
- Network Layer in the Internet

INTERNETWORKING

- **Internetworking**
  - Connecting different networks with different protocols such that a computer on one network can access the data on another network.



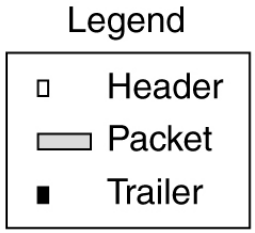
INTERNETWORKING: NETWORK DIFFERENCES

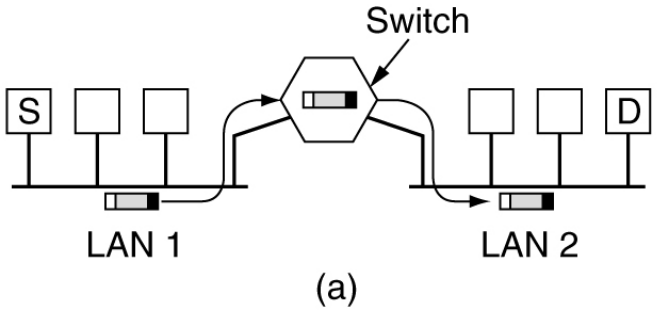
- **How network differ?**

Item	Some Possibilities
Service offered	Connection oriented versus connectionless
Protocols	IP, IPX, SNA, ATM, MPLS, AppleTalk, etc.
Addressing	Flat (802) versus hierarchical (IP)
Multicasting	Present or absent (also broadcasting)
Packet size	Every network has its own maximum
Quality of service	Present or absent; many different kinds
Error handling	Reliable, ordered, and unordered delivery
Flow control	Sliding window, rate control, other, or none
Congestion control	Leaky bucket, token bucket, RED, choke packets, etc.
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, by packet, by byte, or not at all

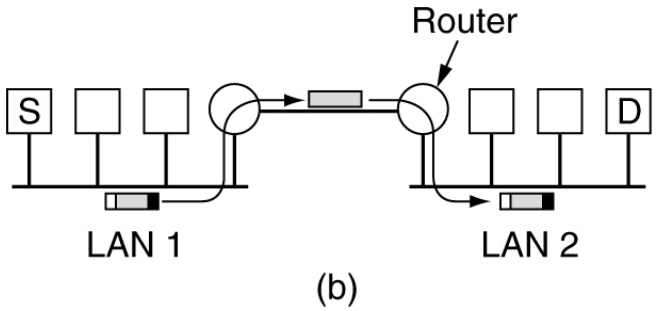
INTERNETWORK: CONNECTION

- **How networks can be connected**
  - Physical layer: repeater, Hub
  - Data Link layer: switch, bridge
  - Network layer: router (gateway: multi-protocol router)
  - Transport layer: transport gateway
  - Application layer: application gateway





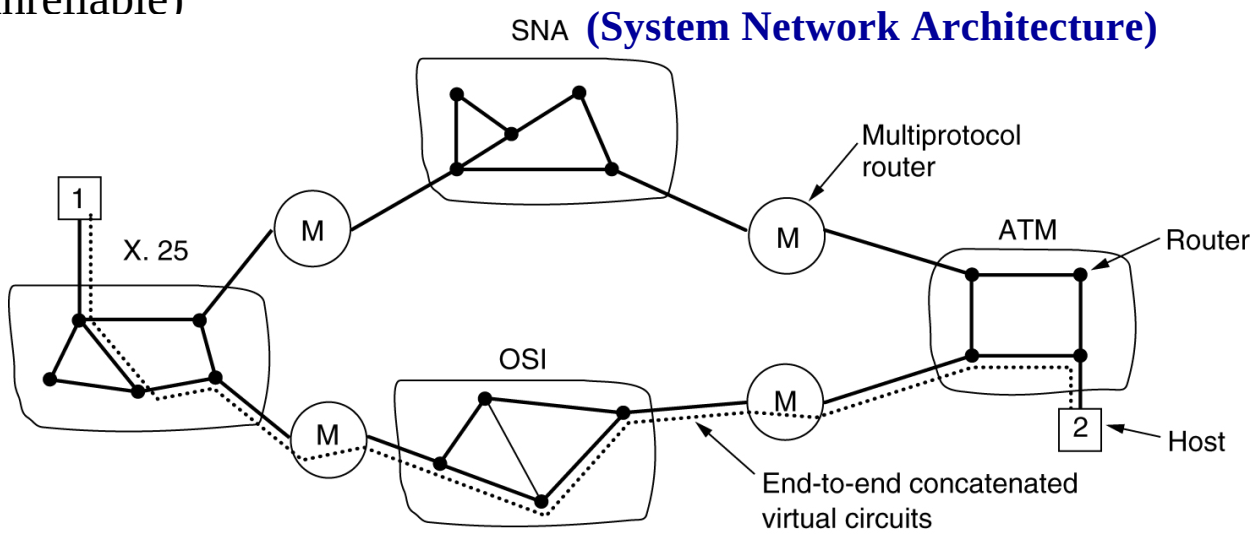
Switch (the entire frame is transmitted)



Router (only IP packet is transmitted)

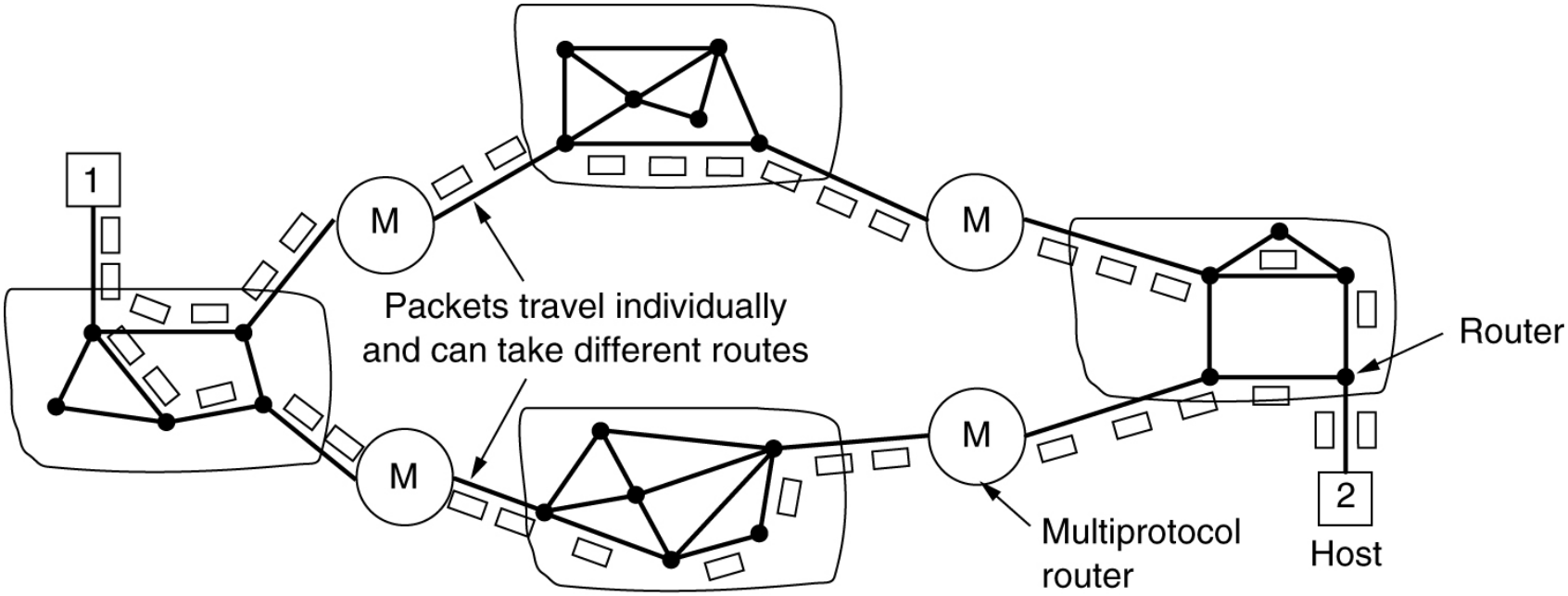
INTERNETWORK: CONCATENATED VC

- **Concatenated Virtual Circuits**
  - Interconnection of networks with connection oriented network layers.
  - Source builds a VC with the gateway (multi-protocol router) that is closest to destination.
  - Gateway builds VCs with intermediate gateways.
  - The last gateway builds a VC with destination.
  - All packets follow the same route.
  - Problem might arise if they support different services (reliable v.s. unreliable)



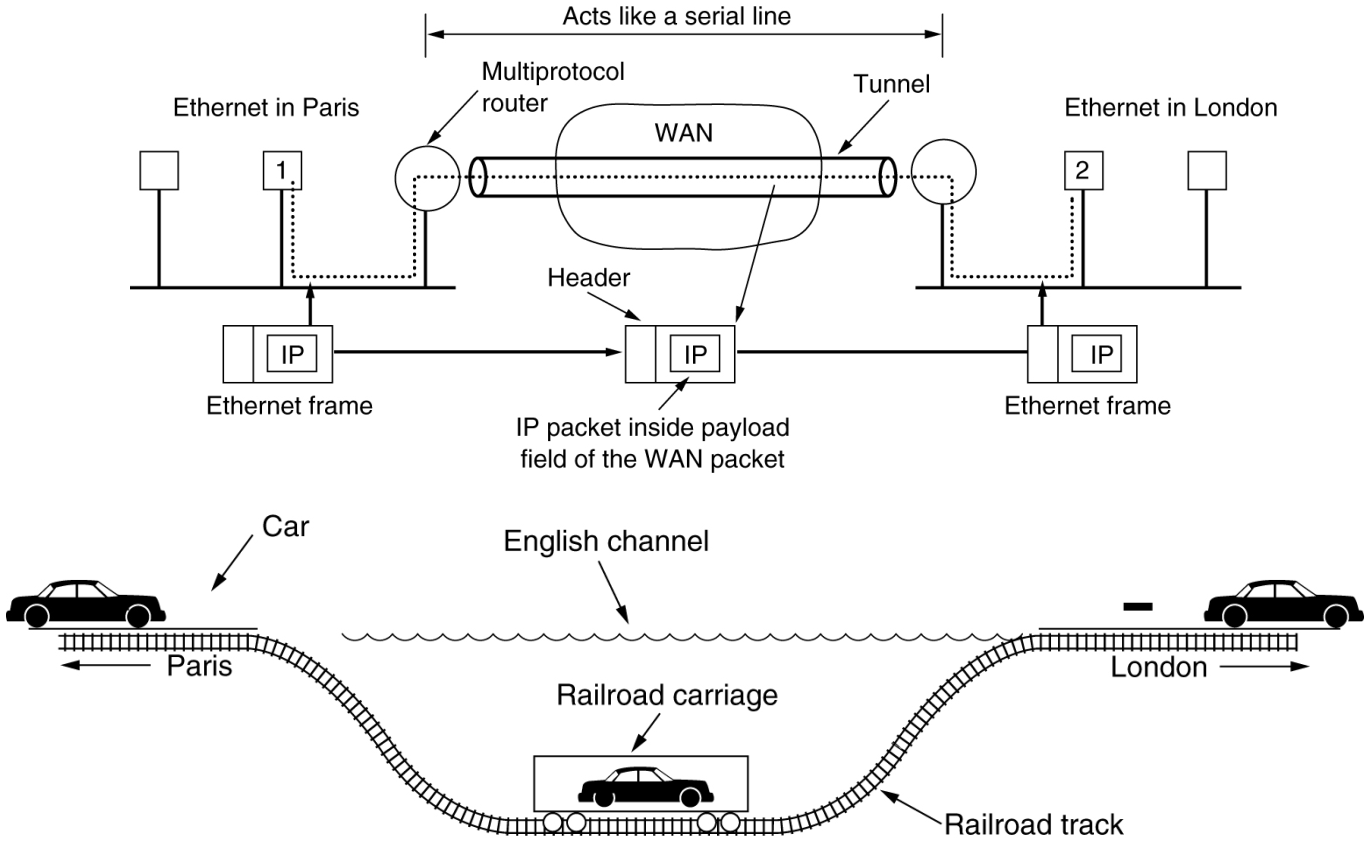
INTERNETWORK: CONNECTIONLESS INTERNETWORK

- **Connectionless Internetwork**
  - Packets might go through different paths.
  - Protocol translation.
  - Address mapping
  - IP packet is a “universal” packet that recognized by most networks.



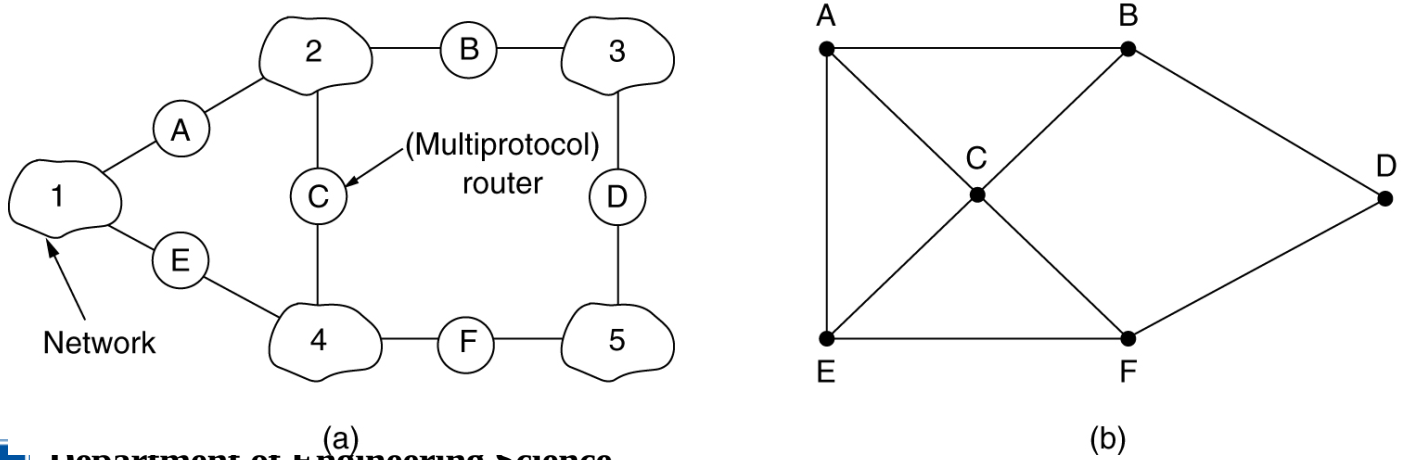
INTERNETWORK: TUNNELING

- **Tunneling**
  - Can be used when source and destination are on same type of network, but a different network is in the middle.



INTERNETWORK: INTERNETWORK ROUTING

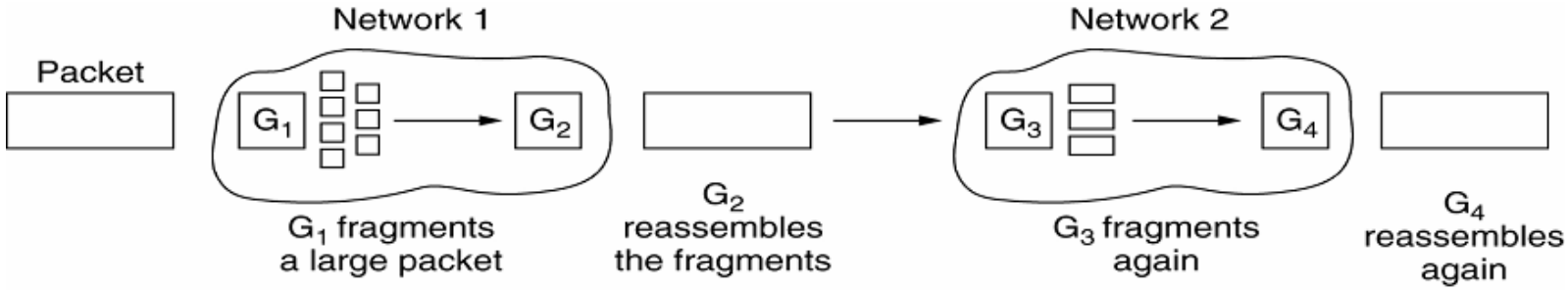
- **Internetwork routing**
  - A packet addressed to a destination at another network is first sent to the local multiprotocol router (gateway).
    - Usually through the MAC address
  - The gateway decides where to forward the packet based on routing table.
    - If same network layer protocol, packet is forwarded directly.
    - If different network layer protocol, tunneling or protocol translation is employed.



(a) (b)

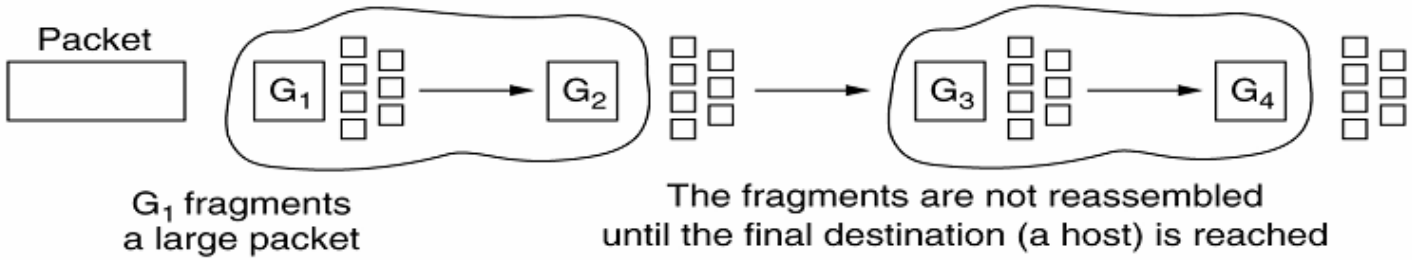
INTERNETWORK: FRAGMENTATION

- **Why fragmentation**
  - different networks have different packet size limitations.
- **Transparent fragmentation**
  - Fragmentation/reassembling are performed at intermediate gateways.



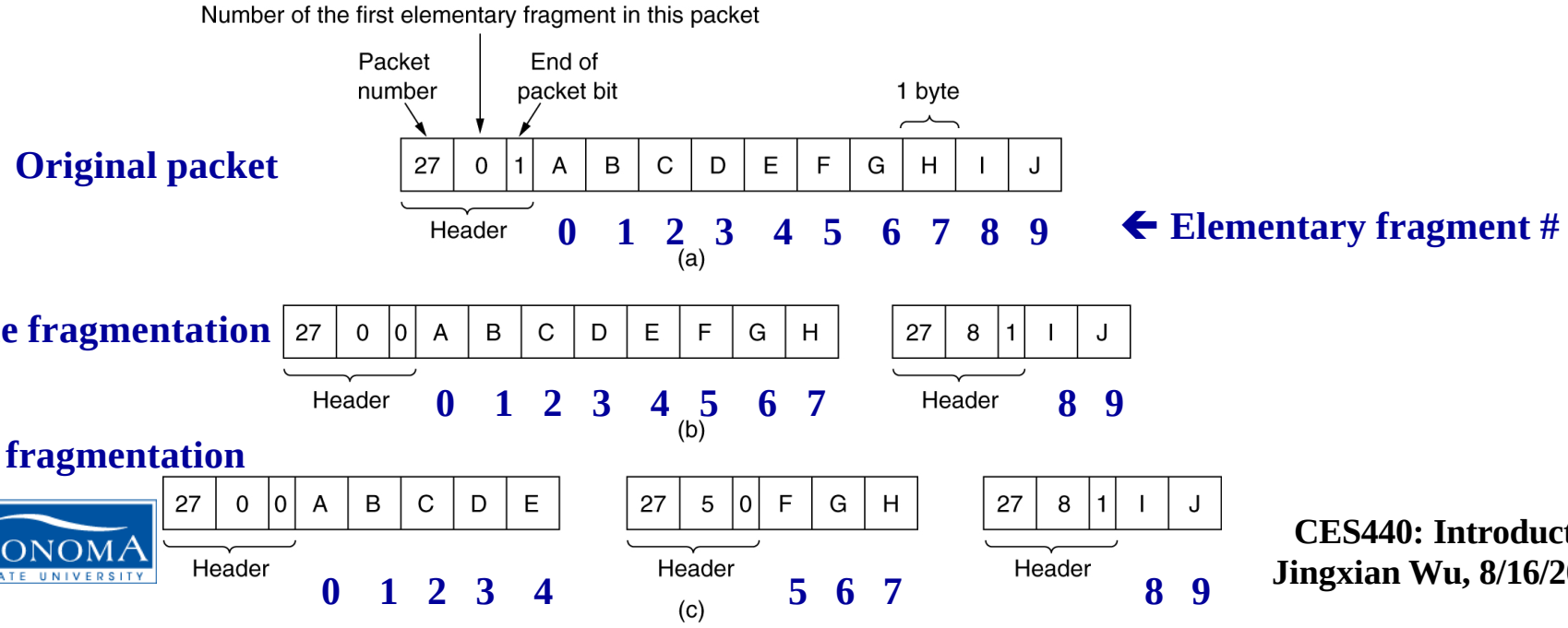
- **Non-transparent fragmentation**
  - Reassembling are performed at destination host.

(a)



INTERNETWORK: FRAGMENTATION

- **How to number the fragment?**
  - Define an elementary fragment size small enough for all networks (e.g. 1 byte)
  - Each fragmentized packet contains several elementary fragments.
  - Each fragmentized packet must have the following three fields
    - Packet # (the original packet #)
    - Elementary fragment # (the seq. # of the first elementary fragment in this packet).
    - End of packet bit (to indicate if this is the last fragmentized packet of the original packet.



OUTLINE

- Design Issues
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- Internetworking
- **Quality of Service**
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QUALITY OF SERVICE (QOS)

- **Flow: a stream of packets from source to destination.**
  - Connection oriented (VC): packets of one flow follow the same route.
  - Connectionless: packet of one flow may follow different route.
  - QoS is defined for each flow.
- **Parameters defining QoS**
  - Reliability
    - Low bit error rate
  - Delay
    - Low transmission delay
  - Jitter
    - Packets arriving at uniform intervals
  - Bandwidth
    - Large bandwidth ➔ High data rate.

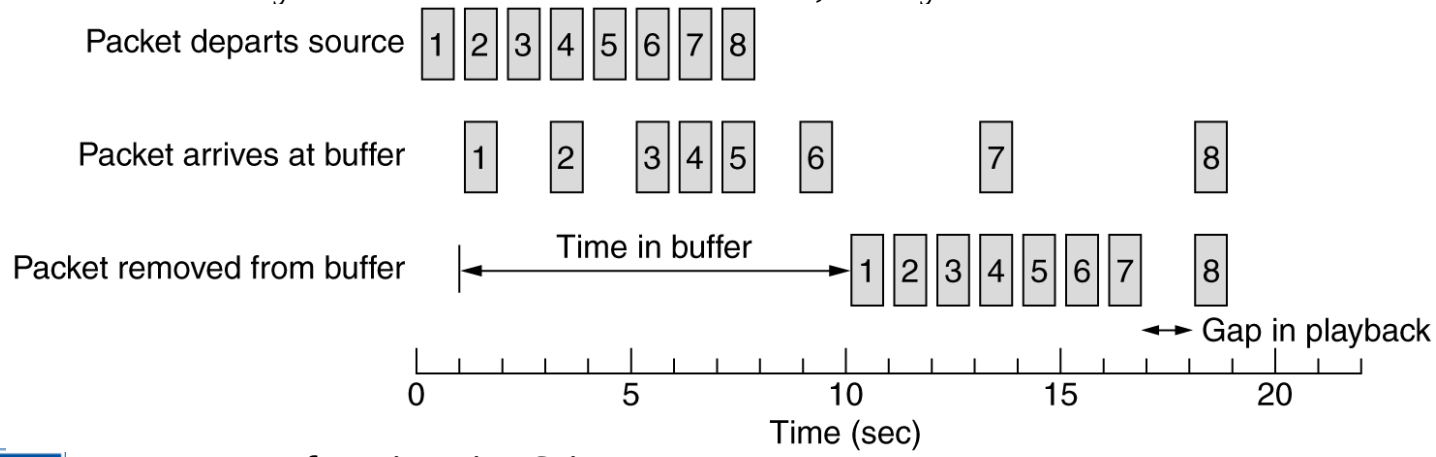
QOS: QOS OF DIFFERENT APPLICATIONS

Application	Reliability	Delay	Jitter	Bandwidth
E-mail	High	Low	Low	Low
File transfer	High	Low	Low	Medium
Web access	High	Medium	Low	Medium
Remote login	High	Medium	Medium	Low
Audio on demand	Low	Low	High	Medium
Video on demand	Low	Low	High	High
Telephony	Low	High	High	Low
Videoconferencing	Low	High	High	High

- **Services defined in ATM**
  - Constant bit rate (telephony)
  - Real-time variable bit rate (compressed videoconferencing)
  - Non-real-time variable bit rate (video on demand)
  - Available bit rate (file transfer)

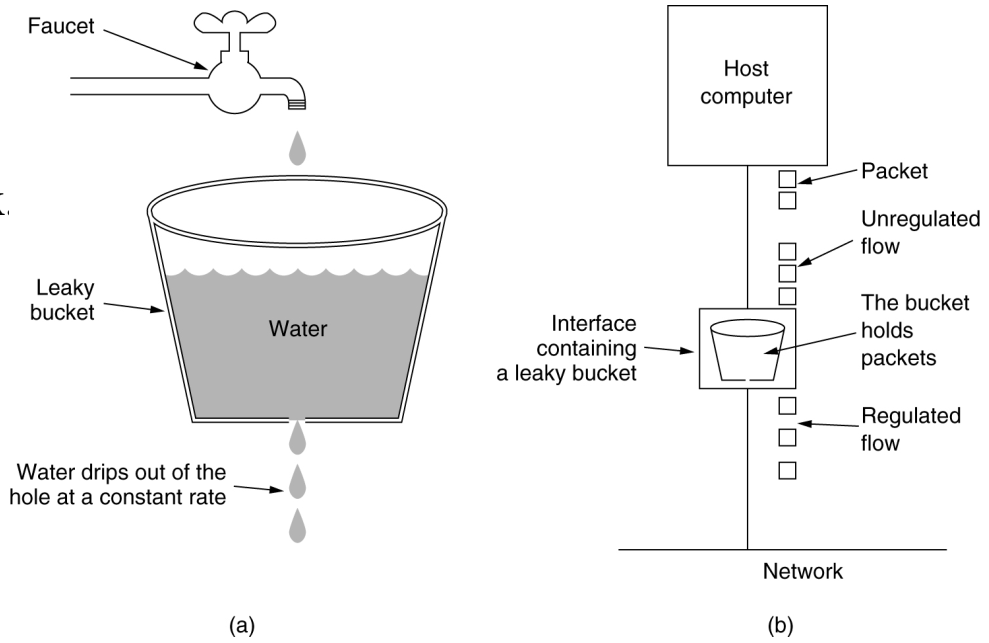
QOS: TECHNIQUES FOR ACHIEVING QOS

- **Overprovisioning**
  - Provide enough resources (router capacity, buffer, bandwidth, etc.) such that all packets can be efficiently delivered.
  - Disadvantage: expensive.
  - Example: telephone system.
- **Buffering**
  - Flows are buffered at the receiver before delivered to higher layer applications.
  - Smooths out jitter.
  - Reliability & bandwidth: not affected; delay: increased.



QOS: TECHNIQUES FOR ACHIEVING QOS

- **Traffic shaping**
  - Smooth out the traffic on the transmitter side: regulating the average rate of data.
  - Service level agreement: during connection setup, user and subnet agrees on certain traffic pattern (e.g. data rate)
  - As long as user carries on its promise, the carrier will deliver it in a timely manner.
- **Leaky bucket algorithm**
  - The host is allowed to put one packet (or certain # of bytes) to network per clock tick.
  - Leaky bucket can be achieved by using a queue at host.
  - If the queue is full, new packet will be discarded.



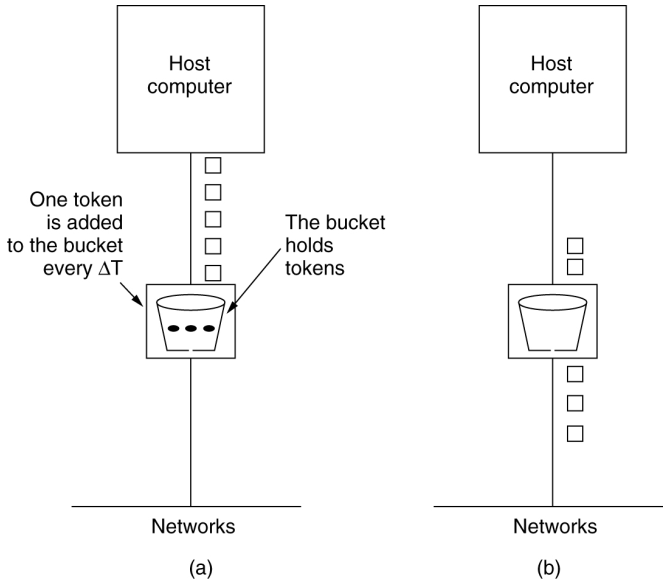
QOS: TECHNIQUES FOR ACHIEVING QOS

- **Leaky bucket (Cont'd)**
  - System with fixed size packets (ATM): put one packet per clock tick.
  - System with variable size packets: (byte counting leaky bucket)
    - At each clock tick, a counter is initialized to *n* (must be larger than maximum packet size)
    - If *packet size* < *n*, packet is transmitted, *n* = *n* - *packet size*
    - Additional packets can be sent as long as packet size < *n*.

      n=*n* - *packet size*
    - When *packet size* > *n*, wait for next tick. (the remaining *n* will not roll-over to next tick!).
  - Enforces a rigid output pattern no matter how bursty the traffic is.
    - Can we speed up the transmission when large burst arrives? ➔ token bucket.

QOS: TECHNIQUES FOR ACHIEVING QOS

- **Token bucket**
  - Bucket holds tokens arriving at a rate of one token per clock tick.
  - In order to transmit a packet (or certain amount of bytes), a token must be destroyed.
  - If no traffic, tokens will be saved for future.
    - Leaky bucket doesn't allow save permission for later use.
  - When the bucket is full of tokens, new token will be discarded.
    - Leaky bucket throw away packets instead.
  - Major difference between Leaky bucket and Token bucket
    - Leaky bucket stores packets in bucket.
    - Token bucket stores tokens in bucket.



QOS: TECHNIQUES FOR ACHIEVING QOS

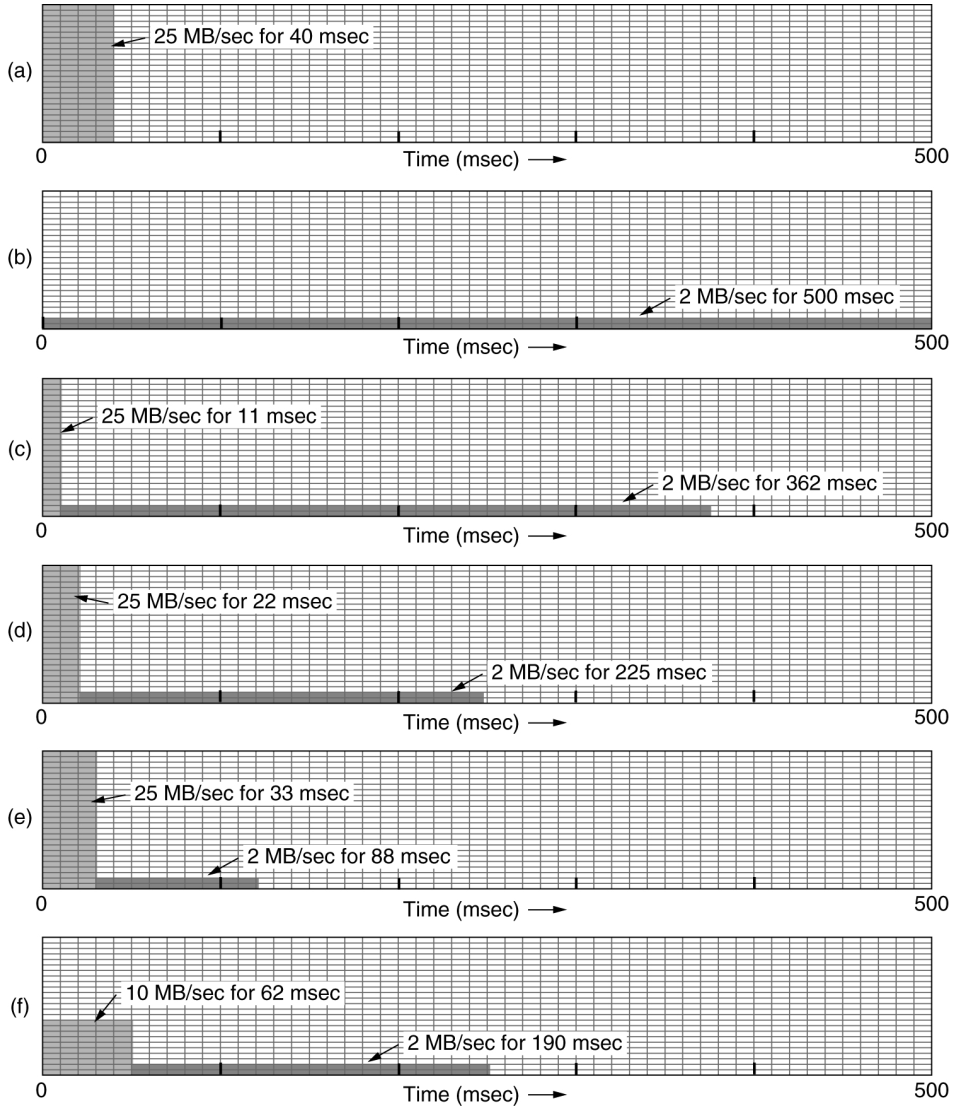
- **Token bucket (Cont'd)**
  - Token bucket capacity *C*=250KB: when a bucket is full, all the tokens allow data of size up to 250KB to be transmitted (e.g. 25 tokens, 10KB/token).
  - Token arrival rate r =2MB/s: tokens arrives at a rate allowing output 2MB/s.
  - Maximum output rate *M*=25MB/s: the maximum output rate supported by the bucket.
  - Example: Given *C*, r, and *M*. How long will the maximum rate burst last?
    - Assume the maximum burst will last *S* seconds.
    - The original amount of data that is allowed by token bucket: C
    - Addition data allowed by token bucket due to arrival of new tokens: rS
    - Total amount of data allowed by token bucket: $B_1 = C + rS$
    - Total amount of data transmitted: $B_2 = MS$

    $B_1 = B_2$ ➔ $C + rS = MS$ ➔ $S = C/(M - r)$

    - For the parameters given above, we have:

$$S = 250\times10^3/(25\times10^6 - 2\times10^6) \approx 11ms$$

QOS: TECHNIQUES FOR ACHIEVING QOS



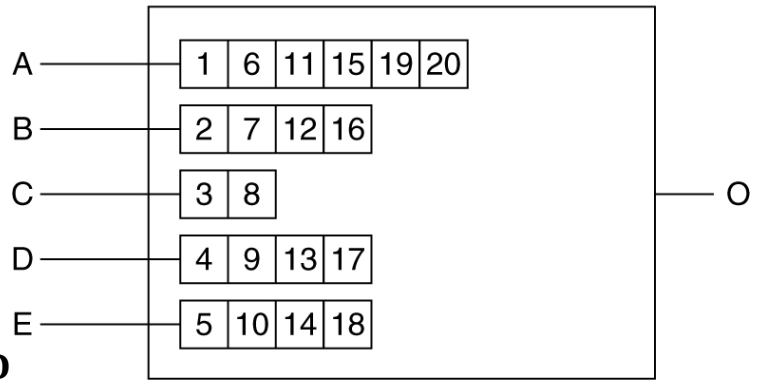
- Data arrival
- Leaky bucket
- Token bucket with C = 250KB
- Token bucket with C = 500KB
- Token bucket with C = 750KB
- Token bucket followed by a leaky bucket

QOS: TECHNIQUES FOR ACHIEVING QOS

- **Resource Reservation:**
  - Reserve the resources along the route of the packet flow.
    - Bandwidth
    - Buffer space: avoid overflow.
    - CPU cycles: make sure packet is processed on time.
- **Admission Control**
  - Routers to decide whether to accept a setup request of a flow based on the available resources: bandwidth, buffer space, CPU cycles.
  - Possible parameters used by user to describe the flow (RFC 2210):
    - 1. Token bucket rate, 2. Token bucket size, 3. Peak data rate, 4. minimum packet size, 5. maximum packet size.
  - Routers convert these 5 parameters into resource requirement (bandwidth, buffer space, CPU cycles) with proprietary algorithms.

QOS: TECHNIQUES FOR ACHIEVING QOS

- **Proportional Routing**
  - Routers usually choose the fastest path ➔ congestion.
  - Divide the traffic into multiple paths in proportion to the capacity of the paths.
- **Packet Scheduling**
  - Motivation: if packets of all flows are stored in same queue, one aggressive sender might use most of the capacity.
  - Solution: maintains one queue for each flow. The queues are served in a round robin fashion: Q1, Q2, …, Qn, Q1, Q2 …
    - Problem:More bandwidth will be used by sender with larger packets.
  - Byte-to-Byte round robin



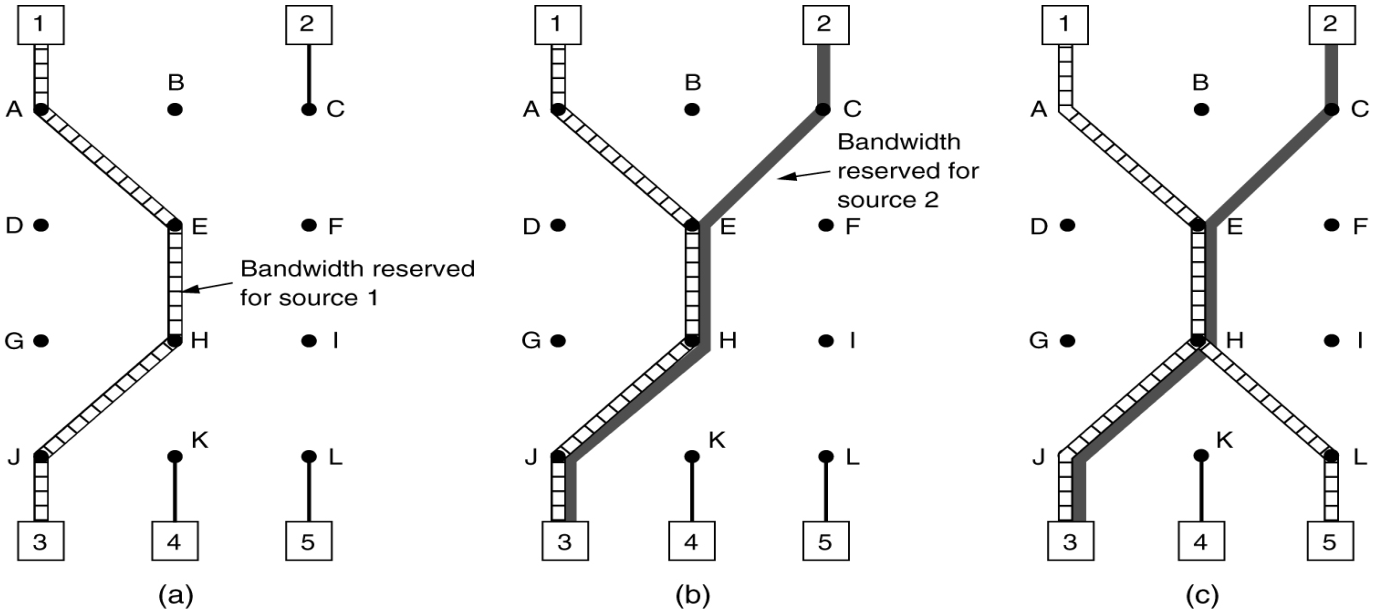
(a)

Packet	Finishing time
C	8
B	16
D	17
E	18
A	20

(b)

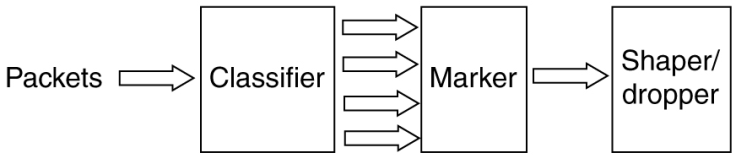
QOS: INTEGRATED SERVICE

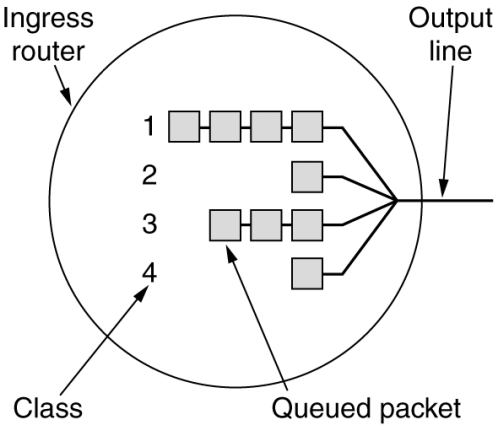
- **Integrated Services with RSVP (Resource Reservation Protocol)**
  - Integrated service: flow based algorithm (How to reserve resource for each flow).
  - Mainly target at multimedia multicasting: Allow users to change channels freely.
  - A spanning tree is built up for each router.
  - Receiver sends request packet back to sender using reverse path forwarding.
    - When routers see the request, resources are reserved.



QOS: DIFFERENTIATED SERVICE

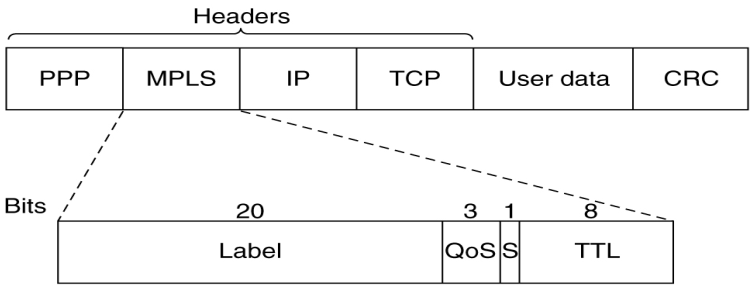
- **Differentiated service**
  - Class-based algorithm: each packet has a *Type of Service* field.
    - Packets belonging to different classes are treated differently.
    - E.g. express mail v.s. regular mail.
- **Method 1: expedited forwarding (RFC 3246)**
  - Expedited packets are sent through a faster path.
  - E.g. 1. more bandwidth allocated to expedited packets.
  - E.g. 2. routers maintains two unfair queues. The queue for expedited packets are served more often.
- **Method 2: assured forwarding (RFC 2597)**
  - 4 priority classes; 3 discard priorities (low, medium, high)➔ 12 service classes.
  - Packets are marked in *Type of Service* field in IP header.





QOS: LABEL SWITCHING AND MPLS

- **MPLS (multiprotocol label switching) (RFC 3031)**
  - Each packet has a label, router forward the packet based on the label. (Virtual Circuit?)
- **Label Switching (LS) v.s. Virtual Circuit (VC)**
  - LS packet still carries destination address. VC packets doesn't.
  - For LS, different flows with the same destination and service class can use one label. For VC, each flow has its own label.
  - LS has no connection. VC is connection orientated.
  - They differ in how the routing table is built.
    - VC: routing table is built during VC setup.
    - LS: two methods: data-driven, control driven.
      - Data driven: the routing table is built with the propagation of the 1st data packet.
      - Control driven: when router boots up, it creates all possible labels based on the hosts directly connected to it. This information is sent to other routers.
- **MPLS packet format**

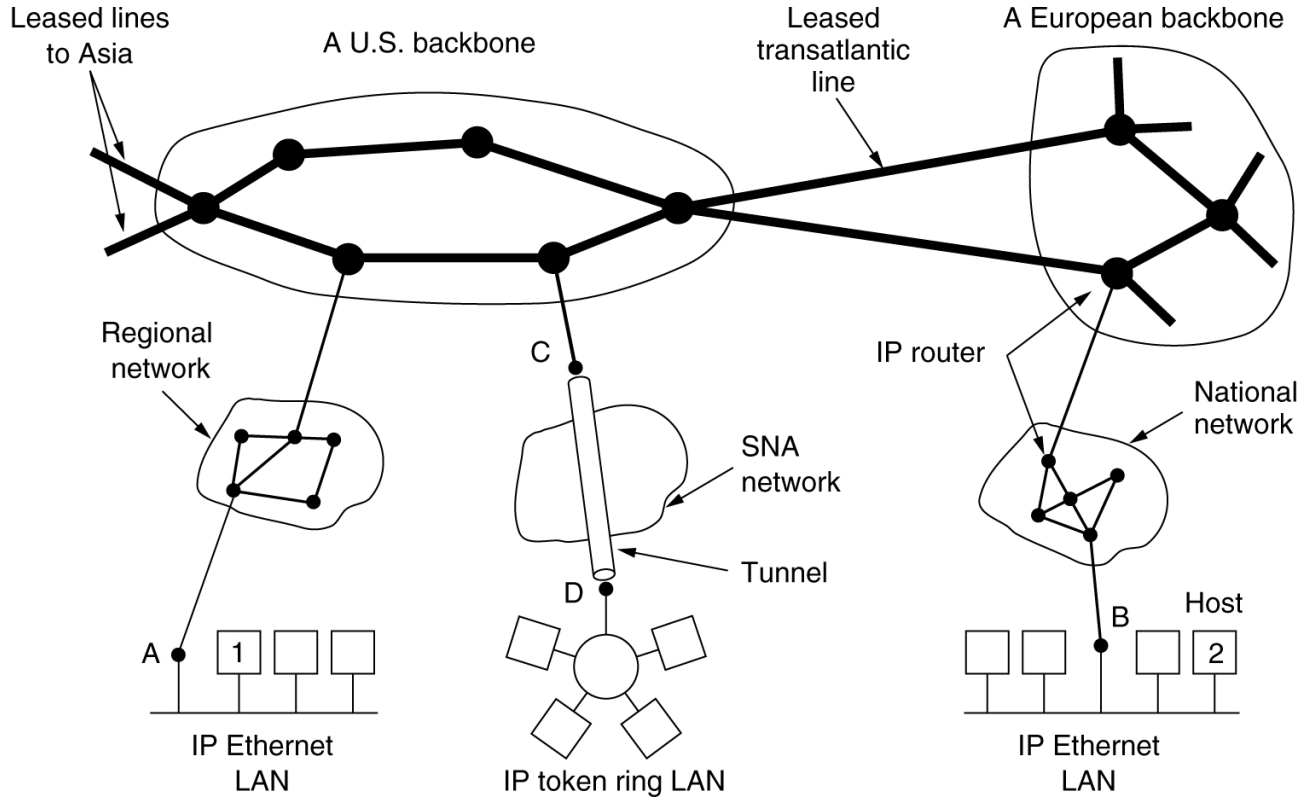


OUTLINE

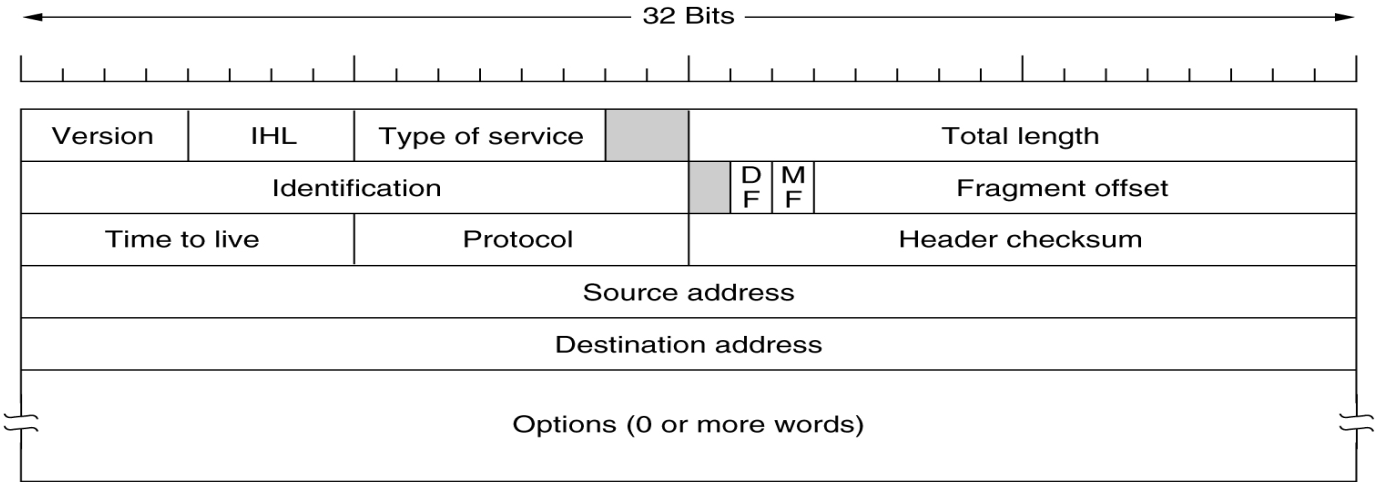
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NETWORK LAYER IN INTERNET

- **Internet Protocol (IP) is the glue for Internet**
  - IP was designed with internetworking in mind.
  - It provides best-efforts (not guaranteed) way to transmit datagram from source to destination.



INTERNET: IP PROTOCOL



- Version: the version of the protocol (IPv4, IPv6)
- IHL (4 bits): header length (in the unit of 32-bit word, or 4-byte word).
  - Min value = 5 ➔ 5 x 4 = 20 bytes. Max value = 15 ➔ 15 x 4 = 60 bytes.
- Type of service (6 bits): used to distinguish different service classes (QoS).
  - Differentiated services: 4 queuing priorities, 3 discard probabilities, etc.
  - This field has been ignored by routers.
- Total length (16 bits): the total length of the IP packet (including header and data)
  - Maximum length: 2^{16} = 65536 bytes = 64KB

INTERNET: IP PROTOCOL

- Identification (16 bits): identify which datagram a fragment belonging to.
  - Fragments belonging to the same datagram has the same ID field.
- DF: Don't fragment.
- MF: More fragment. More fragments are following.
  - All fragments except the last one have this bit set to 1.
- Fragment Offset (13 bits): the seq. # of the first elementary fragment in the current fragment. Elementary fragment: 8 bytes.
  - 2^{13} = 8192 elementary fragments. ➔ 8192 x 8 = 65536 bytes.
- Time to live (8 bits): decrease by 1 after each hop.
  - When hits 0, packet is discarded and a warning is sent back to source.
- Protocol: which protocol is on the transport layer (TCP, UDP, etc.)
- Header checksum: check header only. MUST be recomputed at each router! Why?
- Source address (32 bits): IP address of source.
- Destination address (32 bits): IP address of destination.

INTERNET: IP PROTOCOL

– Option: variable length. Can contain many options
  - Each option starts with 1 byte word to identify the option
  - Some options

Option	Description
Security	Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp

  - More options can be found at www.iana.org/assignments/ip-parameters

INTERNET: IP ADDRESS

- **IP address: 32-bit**
  - 2^{32} = 4295 million addresses.
  - Format: dotted decimal.
    - Binary: 11000000.00101001.00000110.00010100
    - Hexadecimal: C0.29.06.14
      - Hexadecimal:Binary

        0: 0000; 1: 0001; 2: 0010; 3: 0011

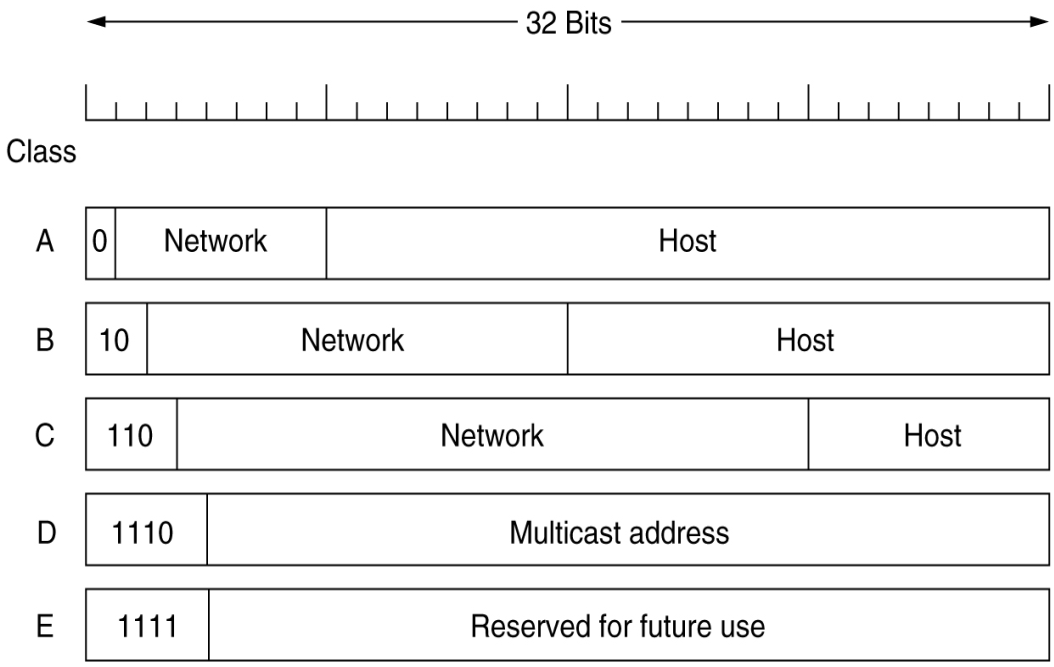
        4: 0100; 5: 0101; 6: 0110; 7: 0111

        8: 1000; 9: 1001; A: 1010; B: 1011

        C: 1100; D: 1101; E: 1110; F: 1111
    - Decimal: 192.41.6.20
      - 12x16+0 = 192; 2 x 16+9=41; 0x16+6=6; 1x16+4=20.
  - IP address assignment is managed by ICANN (Internet Corporation for Assigned Names and Numbers)

INTERNET: IP ADDRESS

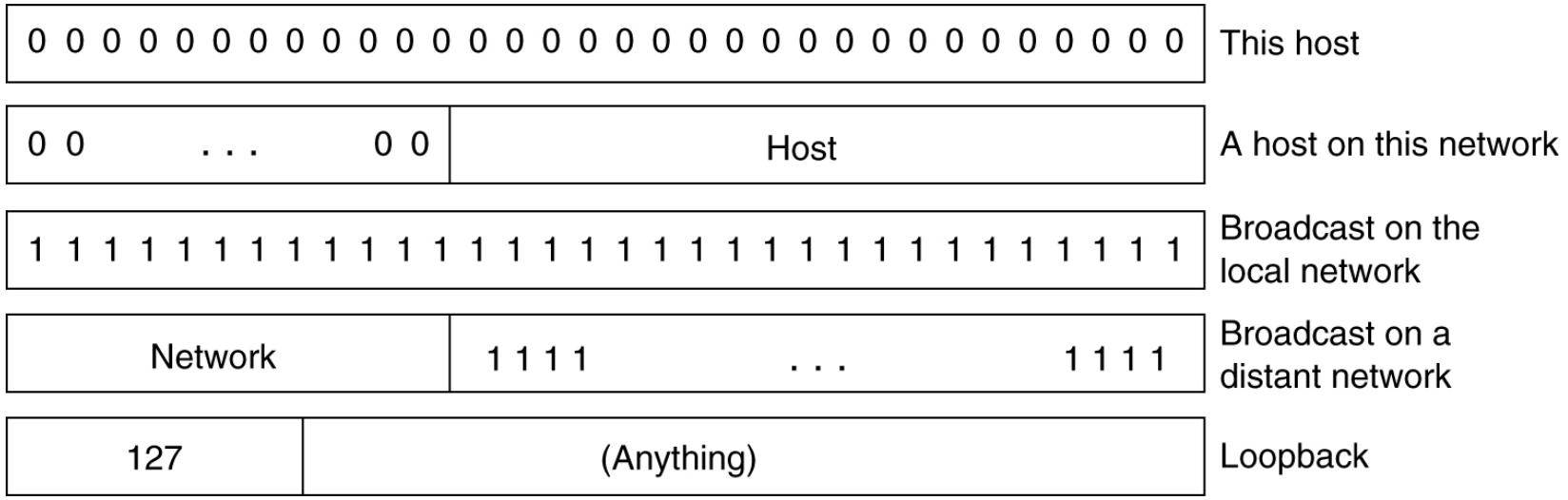
- **IP addresses are divided into 5 classes**
  - 0.0.0.0: this host
  - 255.255.255.255: broadcast in LAN



- 128 networks with 16 million hosts
- 16384 networks with 64K hosts.
- 2 million networks with 256 hosts.
- Multicast
- Reserved

INTERNET: IP ADDRESS

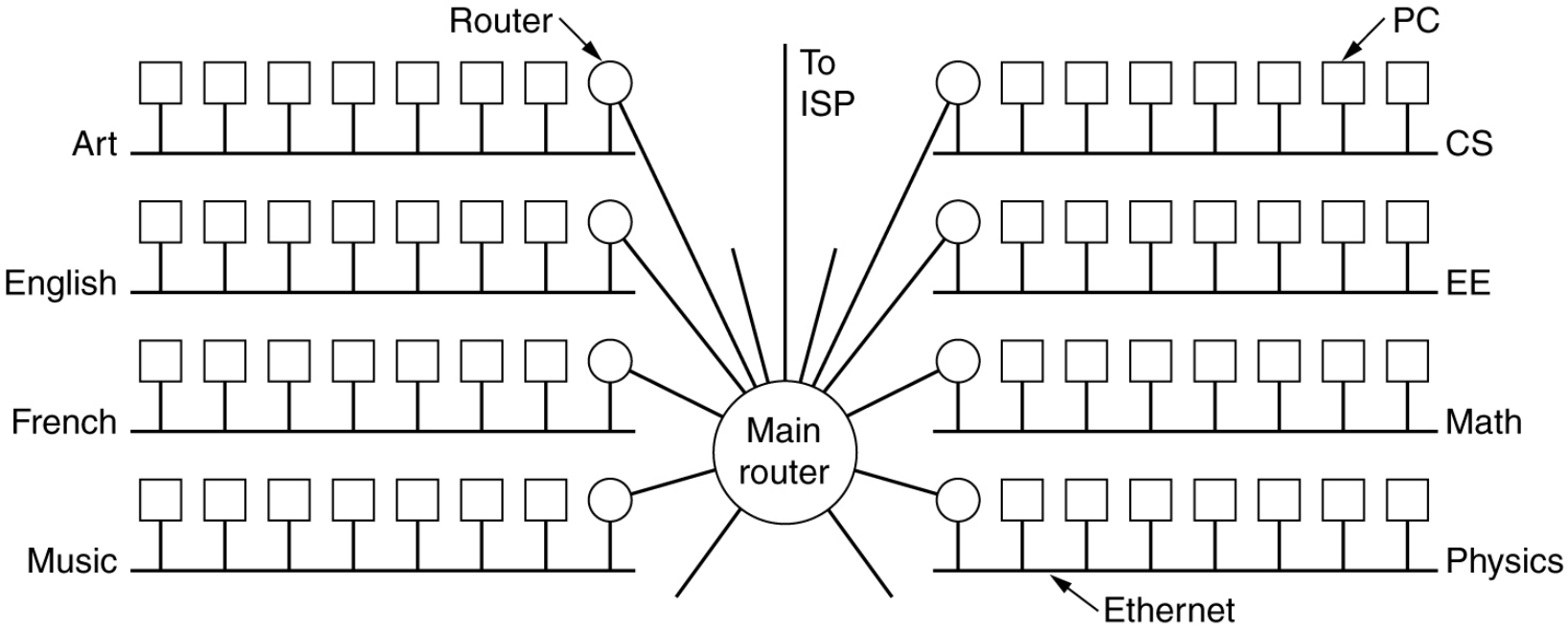
- **Special IP addresses**



  - Loopback: for testing only.
    - Packets sent to this address will not be put in the wire.
    - They are going to be processed locally as incoming packets.

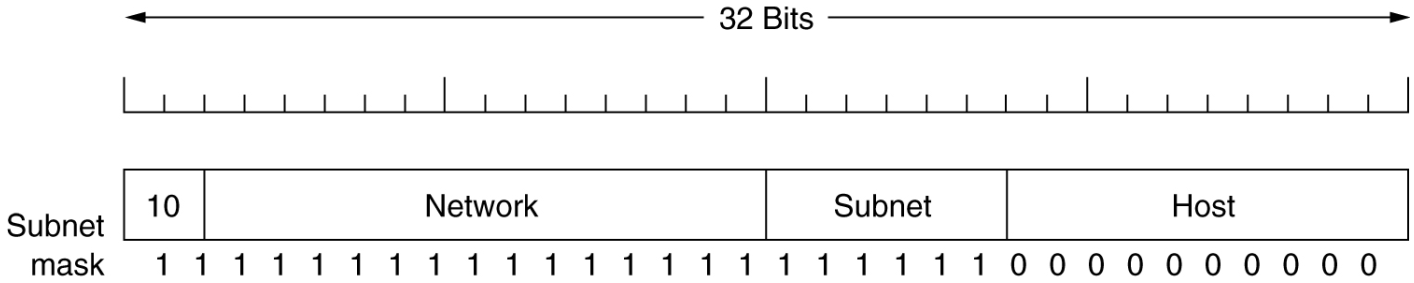
INTERNET: IP ADDRESS

- **Subnet (Different from the collection of routers and Tx Lines!)**
  - Allow a single network be split into small parts for internal use but still act like a single network to the outside.
    - E.g. Each LAN within the network is a subnet.



INTERNET: IP ADDRESS

- **Subnet (Cont'd)**
  - Some bits of the host number are used to identify subnet.
    - E.g. in a class B network, 16-bits are used to identify hosts in the network. Among the 16-bit, 6 bits are used for subnet # (64 Ethernets), 10 bits are used for host # (1022 hosts in each subnet).
    - Subnet mask: 11111111.11111111.11111100.00000000 (255.255.252.0, or 130.157.23.16.0/22)
      - 22 bits are used for network # and subnet #; 10 bits are used for host #
  - Router in the network can route packet based on the subnet #.



E.g: subnet 1: 100000010 00110010 000001|00 00000001
subnet 2: 100000010 00110010 000010|00 00000001

Each IP contains three fields: network address, subnet address, host address